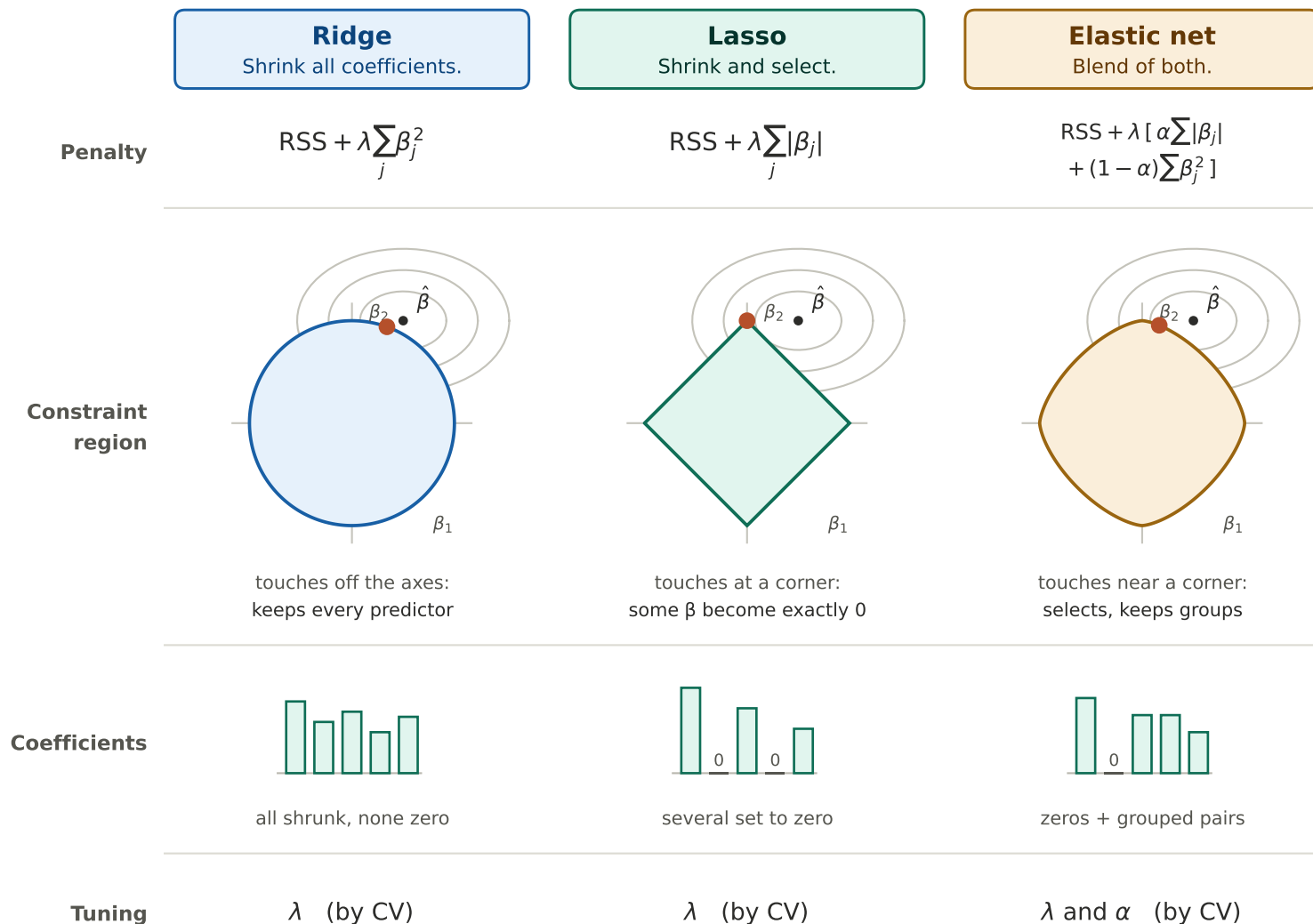


Shrinkage: ridge, lasso, and elastic net

Penalize large coefficients when you fit $\beta_1, \dots, \beta_p$.

Setup

Fit β to minimize RSS + a penalty. The size $\lambda \geq 0$ sets the strength; a larger λ shrinks more. Standardize the predictors first, so the penalty treats them fairly. Choose λ by cross-validation.



Then choose Tune λ (and α for elastic net) by cross-validation, then refit on all the data.

Why the penalties differ

The square penalty (ridge) has a round region, so the fit lands off the axes: every coefficient stays, just smaller. The absolute penalty (lasso) has sharp corners on the axes, so the fit can hit a corner and set coefficients to zero. Elastic net mixes the two. $\alpha = 1$ is lasso, $\alpha = 0$ is ridge. It selects like lasso but keeps correlated predictors together. *More shrinkage adds bias but cuts variance. Cross-validation picks the balance.*

 allowed coefficients (the constraint)

 equal-RSS contours around β

 penalized solution

λ = penalty strength · α = lasso/ridge mix (elastic net)